Project 2
Weather Predictor

Welcome Back to ML... Brother

10.22.2026

The Data

Open-Meteo API: free, no key. Daily Durham weather from 2000 to now, about 9,800 days

Train on 2000–2021, test on 2022–2026. Split by date, not randomly, since two days in a row are almost the same

24 features: today's weather, the high 1, 2, 3 and 7 days ago, 3 and 7 day averages, pressure change

Day of year is turned into sin/cos so Dec 31 and Jan 1 end up next to each other

durham_daily.csv

date	high	low	rain	pressure
2026-10-03	73.0	59.5	0.28	1020.0
2026-10-04	68.5	57.8	0.36	1020.2
2026-10-05	73.8	60.0	0.00	1018.6
2026-10-06	70.1	54.5	0.00	1020.6
2026-10-07	74.5	48.6	0.00	1017.1

Models

baseline

Persistence

every day next week = today's high

baseline

Climatology

the normal high for that date

ML

Ridge

one formula, weight × each feature added up

ML

Gradient Boosting

200 small trees, each one fixes the last one's mistakes

Each ML model is really 7 models, one for each day ahead

Tuned gradient boosting with a grid search (16 settings) using TimeSeriesSplit, which always tests on years after the ones it trained on

Results

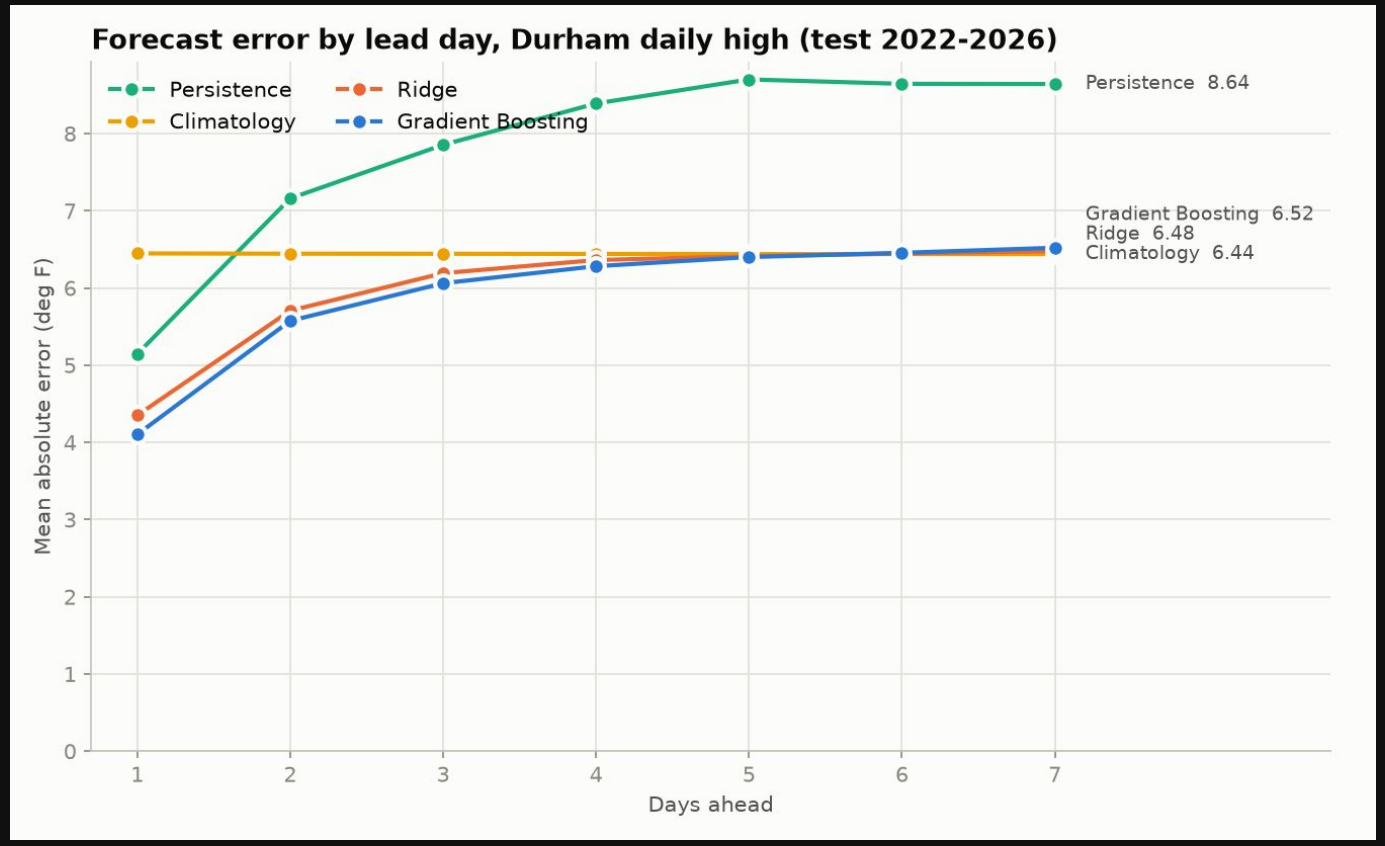
Error = MAE, the average degrees off.

Used this since it predicts a number, not yes/no

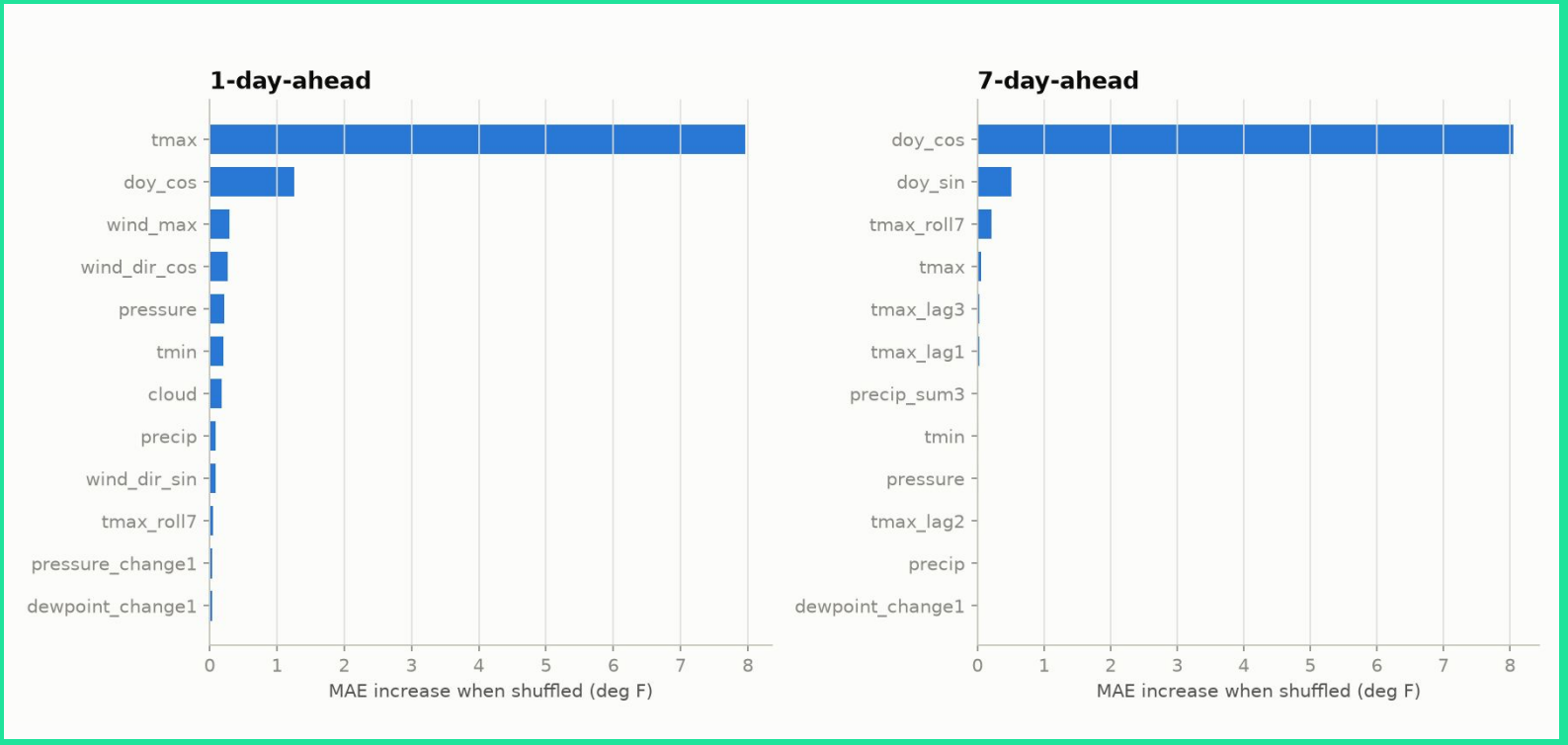
Tomorrow: gradient boosting is off by 4.1°, climatology by 6.4°

Persistence is good for tomorrow, worst by day 2

After about 5 days the ML models are no better than climatology



What the Model Uses



Tomorrow: mostly today's high

7 days out: mostly the day of the year

It figured out by itself that today doesn't say much about next week

Live Test

Ran --predict on Oct 13. Durham had a heat wave, almost 20° above normal

High (°F)	Wed	Thu	Fri	Sat	Sun	Avg off
Actual	90.8	88.6	87.7	73.5	60.4	0.0
My model	82.4	75.9	71.8	68.0	69.0	10.2
Official forecast	91.1	89.0	87.8	69.9	64.9	1.8
Normal high	71.7	71.4	71.2	70.9	70.6	13.1

My model only sees Durham's weather, so it couldn't see the warm air coming. Next step would be adding cities to the west and south

Code demonstration + unique aspects

Resources: Open-Meteo API docs, scikit-learn docs, Claude (documented in README)

Questions?